



Meridian Junior College
JC2 Preliminary
Examinations

Name:

Class:

Index No:

BIOLOGY

9747/03

Paper 3

12 Sep 2008

1 hour 30 minutes

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and register number on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of examination,

1. Write your name, index number and civics group on all sheets of writing papers.
2. Fasten all your writing papers together.

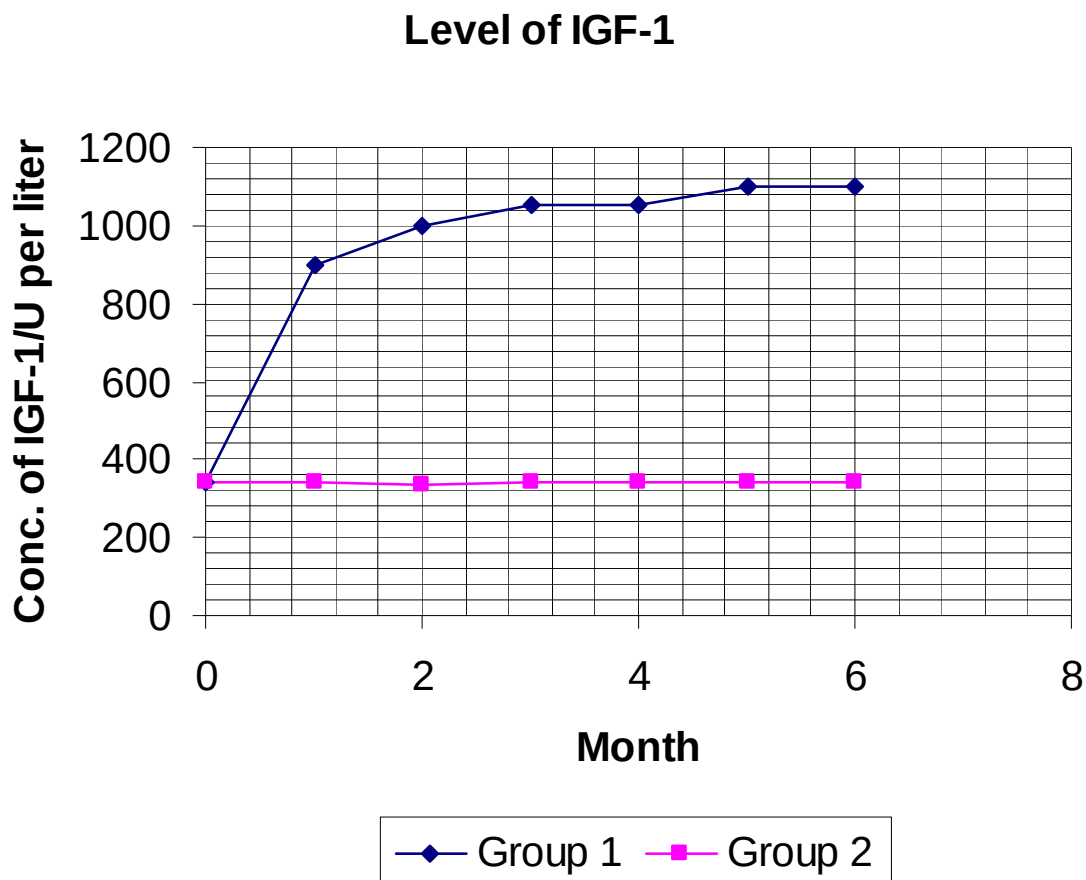
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
Total	

1. In a study on the effect of Human Growth Hormone (HGH) in aging men over 60 years old, the level of growth hormone-insulin-like growth factor 1 (IGF-1) in the plasma of 21 healthy men from 61 to 81 years old were studied during a six-month base-line period and a six-month treatment period that followed. The men all had plasma IGF-1 concentrations of less than 350 U per liter.

Growth hormone secretion can be measured indirectly, however, by measuring the plasma concentration of IGF-I, which is produced and released by the liver in response to growth hormone.

During the treatment period, 12 men (group 1) received approximately 0.03 mg of biosynthetic HGH per kilogram of body weight subcutaneously three times a week, and 9 men (group 2) received no treatment. Plasma IGF-1 levels were measured monthly. At the end of each period, the lean body mass and mass of adipose tissue were measured. The results were shown in Graph 1.1 and Table 1.2 below.



Graph 1.1

Month		0	1	2	3	4	5	6
Group 1	Accumulative % change in lean body mass	0	+2.6	+3.0	+4.7	+6.2	+7.2	+8.8
	Accumulative % change in adipose tissue mass	0	-1.4	-2.9	-4.5	-8.0	-11.7	-14.4
Group 2	Accumulative % change in lean body mass	0	0	0	0	0	0	0
	Accumulative % change in adipose tissue mass	0	0	0	0	0	0	0

Table 1.2

(a) Comment on the effect of human growth hormone on the aging men. [3]

- HGH increased the production of plasma IGF-1, as seen from the increase in IGF-1 concentration from 340 to 1100 U per liter from month 0 to 6/ by the 6th month,
- HGH increased the lean body mass, as seen from the increase by 8.8% by the 6th month.
- HGH decreased the mass of adipose tissue, as seen from the decrease by 14.4% by the 6th month.

The results suggested that the atrophy of the lean body mass and the enlargement of the mass of adipose tissue that are characteristic of the elderly resulted at least in part from diminished secretion of growth hormone. If so, the age-related changes in body composition should be correctable in part by the administration of human growth hormone, now readily available as a biosynthetic product.

(b) Suggest how recombinant HGH plasmids can be produced via genetic engineering. [4]

- **Processed mRNA of HGH gene is extracted from human cells, and reverse transcribed into cDNA by reverse transcriptase.**
- **Bacterial plasmids are cleaved with restriction enzyme to generate blunt ends / sticky ends.**
- **Complementary artificial sticky ends are added to the cleaved plasmids and cDNA / linker DNA is added to the cDNA and cleaved with the same restriction enzyme.**
- **Mix the cleaved plasmids and cDNA to enable complementary base-pairing via hydrogen bonds between the sticky ends to form the recombinant plasmids.**
- **Add DNA ligase to seal the DNA strands by catalyzing the formation of phosphodiester bonds.**

[Max 4]

(c) The recombinant HGH plasmids are inserted into bacteria host cells via transformation. Describe the transformation steps. [2]

- Add **calcium chloride to bacteria cells** to make them competent.
- Subject the **bacteria cells and recombinant plasmids/ ligation mixture** to a brief **heat shock**.

The transformed bacteria were cultured on an agar plate. The plasmid pBR322 was used for the production of recombinant plasmids. There are two antibiotic resistance genes in the plasmid, coding for ampicillin resistance and tetracycline resistance. The restriction enzyme used for the cleaving of the plasmid was AluI and would cleave at the restriction site as shown in Fig 1.3.

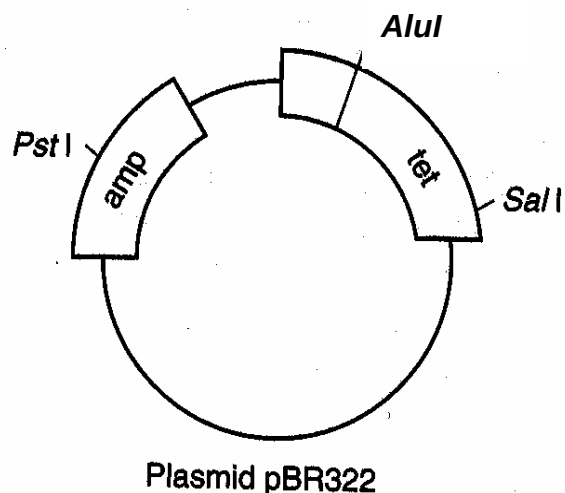


Fig 1.3

(d) Suggest the selection medium/media to use in the agar plate(s) for selection of the bacteria with recombinant plasmids. [1]

- **One** agar plate to contain **ampicillin** and the **other** to contain **tetracycline**.

(e) Outline the steps that will be taken to culture the bacteria on the plate(s) for the selection of the bacteria with recombinant plasmids. [3]

- **Inoculate** the transformed **bacteria** mixture on the **ampicillin plate**.
- **Incubate** the agar plate **overnight** at 37°C.
- **Replica-plate** the bacterial colonies on the **ampicillin plate onto the tetracycline plate**. Incubate the agar plate overnight at 37°C.
- **Compare** the two plates and **pick out** the recombinant bacteria colonies **present on the ampicillin plate but absent on the tetracycline plate and cultured**.

[Max 3]

(f) Suggest what can be done to ascertain the gene in the transformed bacteria with recombinant plasmid is HGH gene? [2]

- Use a **radioactive labeled gene probe** with a **complementary sequence to HGH gene**,

- followed by **autoradiography**.

OR

- **Gene sequencing** is carried out to identify the DNA sequence,
- and matched to known gene sequence through **Bioinformatics**.

OR

- The gene is cloned into an expression vector and **expressed into proteins**,
- which are then identified through **testing for its activity / immunoassay**.

[Total: 15 marks]

2. Cord blood is the blood that circulates through the umbilical cord from the foetus to the placenta. The umbilical cord that remains attached to the placenta has been found to be rich in blood stem cells. Blood stem cells are the young or immature cells that can transform into other forms of essential blood cell types, such as red blood cells, white blood cells and platelets.

Traditionally, the main source of blood stem cells still comes from bone marrow. However, extraction of cells from bone marrow requires a surgical procedure that can be painful and may carry (although minimal) risk to the donor. Cord blood cells, on the other hand, are much easier to collect through a simple procedure. Blood is drawn from the umbilical vein, which takes place after birth and after the umbilical cord is separated from the newborn baby. Therefore, simple cord blood collection procedure involves no risk to the baby or mother.

HLA, or histo-compatibility antigens, which represent immune genotypes, are present on the surface of all cells. The foetus and mother will be of different HLA type. Yet, the foetal cells are immune tolerated by its mother. HLA typing for histo-compatibility testing is used for determining the compatibility of the stem cells (whether bone marrow or cord blood stem cells) of the donor for the recipient. Because of its immunological tolerance, cord blood stem cells can allow for a lower degree of HLA matching as compared to bone marrow (which requires an exact match). Therefore there is a greater chance for a patient to find a match within a cord blood bank compared to a bone marrow registry. There is a decreased risk of graft-vs-host disease for transplantation using cord blood, as compared to using adult bone marrow.

Umbilical cord blood was first used for transplantation in 1988, in France, for a patient with Fanconi's anemia. Since then, cord blood has been increasingly used as a substitute for bone marrow in thousands of transplants worldwide. Several previously life-threatening diseases such as SCID (Severe Combined Immune Deficiency), leukemias, lymphomas, thalassemia major and severe aplastic anemia can now be cured using stem cell transplantation.

- (a) Describe the unique features of all stem cells. [3]
- Stem cells are **unspecialised**, there is an **absence of tissue-specific structures** for specialised functions;
 - Stem cells are capable of **continually renewing and dividing / proliferate / replicate many times** for long periods.
 - Stem cells are capable of **differentiating into specialised cell types** under **appropriate conditions**.
- (b) Explain the advantages of using cord blood stem cells in place of bone marrow. [2]
- It is **not painful**; **blood is drawn** from umbilical vein **after umbilical cord is separated from the newborn baby / no surgical procedure involved**.
 - **Cord blood stem cells** are **easier to collect**; **blood is drawn** from umbilical vein **after umbilical cord is separated from the newborn baby / no surgical procedure involved**.
 - **There is no risk to the baby or mother**; **blood is drawn** from umbilical vein **after umbilical cord is separated from the newborn baby / no surgical procedure involved**.
 - **Greater chance to find match / donor**; because cord blood stem cells **allow for lower degree of HLA matching** as compared with bone marrow.
 - **Better chance of successful transplantation**; because there is a **decreased risk of graft-vs-host disease**. REJECT: Immunological tolerance.
- [Any 2]
- Cord blood stem cells are regarded as multipotent stem cells that have the potential to differentiate into a limited number of cell types. There exists other groups of stem cells of greater developmental potential.
- (c) Describe with an example for each, the two groups of stem cells with higher developmental potential than cord blood stem cells. [2]
- **Totipotent** stem cells retain the complete set of genetic information and have potential to **become any cell type including that of the extraembryonic membranes** e.g. the **fertilized egg / first 4 cells of fertilised egg**.

- **Pluripotent** stem cells retain the complete set of genetic information and have potential to **become any cell type except that of extraembryonic membranes** e.g. **embryonic stem cells / embryonic germ cells / embryonic carcinoma cells.**

Besides using stem cell transplant to treat genetic disorders such as SCID, gene therapy could be another treatment option. Severe Combined Immunodeficiency (SCID) is a disease of young children in which patients have neither cell-mediated immune responses nor the ability to make antibodies. There are two types of SCID: Adenosine Deaminase (ADA) deficient SCID and X-linked SCID. There are also two types of delivery system used in SCID gene therapy: non-viral gene delivery system and viral gene delivery system.

(d) Suggest a suitable viral vector used in SCID gene therapy. [1]

- **Retrovirus / adenovirus / adeno-associated virus**

(e) Outline the use of viral gene delivery system in gene therapy. [3]

- **Viruses are genetically engineered to carry normal human gene.**
- The viruses are used as **vector** to **infect target cells.**
- The **normal gene** may get **incorporated into host genome** / remains **episomal.**

X-linked SCID is caused by a mutated X-linked gene encoding a subunit called the gamma c (γ_c) of the receptor for several interleukins including interleukin-7 (IL-7). IL-7 is essential in converting blood stem cells into precursors of T cells. A non-viral gene delivery system is available to specifically target the mutated gene for the treatment of X-linked SCID.

(f) Describe the non-viral gene delivery system used in the gene therapy of X-linked SCID which specifically targets the mutated gene. [4]

- The non-viral gene delivery treatment for X-linked SCID consists of a synthetic protein containing a **zinc-finger transcription factor** engineered to recognize and **bind to the γ c locus**.
- This protein is **coupled to a restriction enzyme** that **cleaves** both strand of DNA near that locus to create a double-stranded break in the DNA **at the γ c / mutated locus**;
- And a **plasmid containing the correct version of the γ c gene** as a template.
- The protein thus uses its own machinery for the repair of the double stranded break by **homologous recombination**, hence **repairing the defective / mutant gene** with the correct version of the γ c genes.
- Use of **liposome / gene gun / electroporation / microinjection** to deliver zinc finger factor into host cell.

[Total: 15 marks]

3. Plant growth regulators (PGR) such as auxin, cytokinin and ethylene are naturally produced in plants and are important in determination of the developmental pathway of plant cells. Cheaper synthetic analogues are frequently used in plant tissue culture instead of the natural PGR. Table 3.1 summarizes the general characteristics and roles of some natural and synthetic PGR.

Type of PGR	Examples	Role
Auxins	IAA(natural) - unstable to heat & light	Promote cell division and cell growth Root initiation (when auxin:cytokinin is high)
	2,4-D(synthetic) – stable to heat & light	
Cytokinins	2iP(natural) – unstable to heat & light	Promotes cell division Shoot formation (when auxin:cytokinin is low)
	Kinetin(synthetic) - stable to heat & light	
Abscisic acid (ABA)	-	Inhibits cell division Maturation of somatic embryos Abscission of plant leaves Seed dormancy Induces stomatal closure to reduce water loss by transpiration
Ethylene	-	Abscission of leaves and flowers Seed and bud dormancy by growth inhibition Fruit ripening

Table 3.1

Prunus lannesiana is an early-flowering cherry (only in spring) in Japan Izu peninsula. It is commonly known as Sakura or Japanese Cherry. Researchers are keen to propagate *P. lannesiana* from sterilized explants by micropropagation due to the advantages that the technique offers. Hence, a study was made to analyse the concentration of PGR present in the *P. lannesiana* in the four seasons and the results were summarized in Table 3.2.

Table 3.3 illustrates the physiological development of *P. lannesiana* in the various seasons.

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Season Conc. of PGR /arbitrary unit	Spring (Mar-May)	Summer (June – Aug)	Autumn (Sep-Nov)	Winter (Dec – Feb)
IAA (auxin)	15	20	25	30
2iP (cytokinin)	10	15	10	5
ABA	5	5	12	15
Ethylene	12 – 20	18	16	14

Table 3.2

Season	Spring	Summer	Autumn	Winter
Shoot development	Very Active	Active	Minimal	Nil
Root development	Minimal	Moderate	Active	Very Active
Leaves development	Very Active	Active	Senescence and abscission	Nil
Flowers development	Short full bloom in early spring, followed by senescence	Nil	Nil	Buds develop but dormant in late winter
Fruit and seed development	Fruit and seed development in late spring	Seed maturation	Seed dormant	Seed dormant

Table 3.3

(a) State two limitations of micropropagation.

[2]

1. **Contamination** by bacteria can **cause high losses** in a short time.
2. Micropropagation **requires sophisticated facilities, sterile laboratory conditions** and **special nutrient media**.
3. It also **requires trained personnel** with specialised skills / **technical expertise**.
4. **High cost** of micropropagation due to the facilities needed and labour input.
5. **Genetic off-types** may occur when tissue culture is used for cloning plants.

[Any two of the above]

(b) Suggest why synthetic PGR (e.g. 2,4-D and Kinetin) are used instead of natural PGR (e.g. IAA and 2iP) in plant tissue culture.

[1]

1. **Synthetic PGR** are **cheaper**
2. and **more stable to heat and light**.

(Reject: easier to obtain)

[Any one of the above]

(c) (i) Using the information provided in Table 3.2 and Table 3.3, explain the effect of the change in the auxin:cytokinin ratio from Spring to Winter and vice versa. [4]

1. When season change from spring to winter, the **auxin:cytokinin ratio generally increased from 3:2 in spring to 6:1 in winter / four fold;**
2. which **promotes root growth in winter to improve absorption of nutrients and water** to sustain the survival of plant.
3. When season change from winter to spring, the **auxin:cytokinin ratio decreased from 6:1 in winter to 3:2 in spring / four fold;**
4. which **promotes shoot development in spring, allowing for full bloom of *Prunus lannesiana* in spring for pollination / for photosynthesis for maximal growth / for regeneration of leaves for photosynthesis.**

(c) (ii) Using the information provided in Table 3.2 and Table 3.3, explain the effect of a change in concentration of ABA from Spring to Winter. [4]

1. **ABA concentration increases from 5 arbitrary units in spring/summer to 12 arbitrary units in autumn; OR ABA concentration increases by 7 arbitrary units from spring/summer to autumn;**
2. **which promotes the abscission of plant leaves in autumn to reduce water loss / induces stomatal closure to reduce water loss by transpiration in autumn / promote seed dormancy.**
3. **ABA concentration increases further from 12 arbitrary units in autumn to 15 arbitrary units in winter; OR ABA concentration increases by 3 arbitrary units from autumn to winter;**
4. **which maintains seed dormancy in winter.**

A gene encoding ethylene inhibitor protein has been isolated from a related species and researchers would like to clone this gene into *P. lannesiana* in an effort to extend the blooming period. Fig 3.4 illustrates a Ti plasmid which is commonly used to transfer genes into plants using the soil bacterium *Agrobacterium tumefaciens* as a vector.

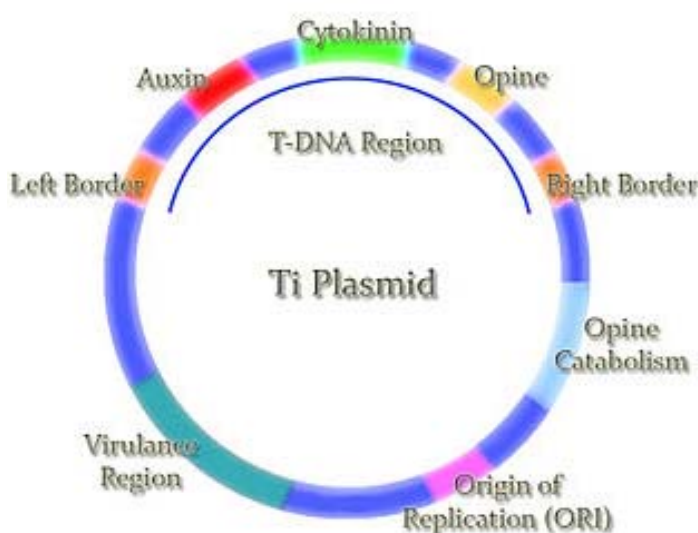


Fig 3.4

(d). Describe how the gene of interest can be inserted into Ti plasmid to form a recombinant Ti plasmid. [4]

1. **Genes encoding the proteins for the production of auxin and cytokinin are removed** from the T-DNA fragment **using restriction enzyme digestion**;
2. **Gene of interest is inserted between the left and right border / in the place of auxin and cytokinin genes / in the T-DNA region.**
3. **Gene of interest and Ti plasmid will anneal by complementary base pairing** between their sticky ends.
4. **DNA ligase seal the gaps** between the gene of interest and the Ti plasmid by catalysing the **formation of phosphodiester bonds** to form the recombinant Ti plasmid.

[Total: 15 marks]

Write your answers to this question on the writing paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc, as indicated by the question.

4. (a) Describe how nucleic acid hybridization can be used in RFLP analysis on isolated DNA sample. [6]

1. **Digest genomic DNA** samples with specific **restriction enzyme** to produce **restriction fragments**.
2. Use **gel electrophoresis** to **separate the DNA fragments** by size/molecular weight.
3. **Southern Blotting** transfer **DNA fragments** from the gel to the sheet of **nitrocellulose filter**.
4. **Sodium hydroxide added** to **nitrocellulose membrane** to denature the DNA fragments into **single stranded DNA**.
5. The DNA fragments on the nitrocellulose membrane **hybridized/ complementary base pair with radioactively labeled/ fluorescent probe** in solution by hydrogen bonding.
6. **Excess probe is washed off the nitrocellulose membrane**.
7. **Visualize the band patterns** containing the DNA that hybridize with radioactive probe by **laying a sheet of photographic film over the nitrocellulose filter / using autoradiography** to yield an image of specific radioactively labeled DNA bands.

OR

Visualize the band patterns containing the DNA that hybridize with fluorescent probe by **UV light**.

- (b) Explain the application of RFLP analysis in disease detection and DNA fingerprinting. [8]

Disease detection

Detection of defective genes:

1. Defective genes that arose as a result of **gene mutation** may be identified using RFLP.
2. Mutation leads to **different number / molecular weight / length of restriction fragments**.
3. The **DNA profile / RFLP** can be **analysed / compared against the normal DNA** (negative control) and **diseased profile** (positive control) to detect the mutation.

Detection of pathogens:

4. The **DNA of known pathogens** can be detected as known pathogens have a **specific RFLP profile**.

Example:

5. E.g. **Huntington disease / Dengue fever / Malaria / Sickle cell anaemia / Haemophilia / Cystic fibrosis / SCID** / other valid disease example (reject: viral disease with high mutation rate).

[max 4 points]

DNA fingerprinting

6. Different DNA fingerprint arose as a result of **different number of tandem repeat** among individuals.
7. Different number of tandem repeat leads to generation of **different number / molecular weight / length of restriction fragments**.
8. The **DNA fingerprint** can be **analysed / compared against another individual** to **detect matching** band(s) / pattern.
9. E.g. **Paternity testing / forensic analysis for crime investigation/ trace evolutionary lineage**.

[max 4 points]

(c) Explain how gel electrophoresis is used to analyse nucleic acids. [6]

1. Gel electrophoresis **separates nucleic acids** on the basis of their rate of movement **through a gel** in an electric field.
2. **Rate of movement** depends on **size** of the **DNA**.

3. The **bigger the size** of the DNA fragment, the **slower** the migration/ the **shorter the distance** migrated through the gel / **vice versa**.
4. **Negatively charged** nucleic acid will migrate **towards the anode**.
5. The **DNA** will be **stained / intercalated with ethidium bromide** and
6. **visualised** under **UV** light.
7. The **bands** will be **compared to the DNA ladder** to **estimate/ calibrate their molecular size**.

[max 6 points]